

AFIF FURQON

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Electrical Engineering graduate with comprehensive expertise in Electrical Power Systems, Renewable Energy, and *Internet of Things* (IoT). Practical track record in power plant control system operations, substation supervision, and IoT *hardware* development (ESP32, LoRaWAN). Proficient in integrating hardware and software engineering through C++ and Python for data acquisition and analysis. Committed to applying automation and instrumentation expertise to support operational reliability and technological innovation in the Energy, Oil & Gas, Mining, and Transmission industries.

SKILLS

Hardware/Embedded:	Microcontrollers (ESP32, Arduino), <i>Programmable Logic Controller</i> (PLC), HMI, LoRaWAN.
Software & Engineering Tools:	C++, Python, TinkerCAD, AutoCAD, Proteus, Arduino IDE, <i>ThingsBoard</i> IoT Platform, <i>Cisco Packet Tracer</i> , GitHub.
Engineering Concepts:	Industrial Automation & SCADA, Instrumentation & Control, Electrical Power System Maintenance, Substation Supervision, <i>IoT System Architecture</i> .
Soft Skills:	Problem Solving (Troubleshooting), Teamwork, Time Management, High Adaptability, Occupational Safety Culture (K3/HSE).

PROFESSIONAL EXPERIENCE

PT. PLN Enjiniring – West Jakarta, DKI Jakarta **Sep 2025 – Jan 2026**

Engineering Internship – Construction Supervision

- Supervised field operations on the Suralaya Unit 3 migration project (transition of transmission and generation systems from 500 kV to 150 kV) with 8 field personnel, achieving a record of 0 work accidents.
- Monitored the dismantling and installation of main substation components, including the 470 MVA Generator Transformer, *Disconnecting Switches* (DS), *Circuit Breakers* (CB), and *Capacitive Voltage Transformers* (CVT).
- Ensured field implementation compliance with *engineering* technical specifications during the replacement of the *grounding* system and conductor installation.
- Monitored decommissioning activities through to the installation of the new 470 MVA generator transformer.

PT. PLN NP PLTU (330 MW) – Indramayu, West Java **Feb 2025 – Mar 2025**

Engineering Internship – Instrumentation and Control Systems

- Performed corrective and preventive maintenance on 5 *coal mill* units, ensuring continuous equipment reliability and optimal performance for 330 MW power plant operations.
- Collaborated with 4 maintenance team members to inspect, *troubleshoot*, and replace 7 critical sensors (4 *pressure sensors* and 3 *limit switches*) in the power plant control system and *coal feeder* mechanism.
- Supported valve replacement activities in the H2 Plant area, strictly complying with Occupational Health and Safety (K3) protocols, thereby contributing to a record of 0 work accidents (Zero LTI).

Smart Electronic Systems Laboratory, Institut Teknologi PLN – West Jakarta, DKI Jakarta

Deputy Coordinator of Digital Engineering Laboratory

Mar 2024 – July 2026

Telecommunication Systems Laboratory Assistant

Sep 2024 – Jan 2025

- Coordinated practicum operations and led a team of lab assistants for the Digital Engineering and Telecommunication Systems modules, facilitating learning for more than 700 bachelor's (S1) and diploma (D3) students.

- Guided laboratory experiment sessions comprehensively, covering logic circuit design, embedded systems, up to telecommunication signal transmission and modulation.
- Guided and assisted practicum participants in using simulation tools such as Cisco Packet Tracer, Proteus, and TinkerCAD to troubleshoot while completing final laboratory projects.

Electrical Research Centre (ERC), Institut Teknologi PLN – West Jakarta, DKI

Mar 2025 – Feb 2026

Jakarta

ERC Staff

- Introduced robotics and drone technology to 50+ new members through basic demonstrations and knowledge-sharing activities.

EDUCATION

Institut Teknologi PLN – West Jakarta, DKI Jakarta

Sep 2022 – Nov 2026

Bachelor's Degree (S1) in Electrical Engineering | GPA 3.74/4.00

ENGINEERING PROJECTS

Institut Teknologi PLN – West Jakarta, DKI Jakarta

Sep 2022 – Nov 2026

Design and Development of an IoT-Based Early Fire Detection System Prototype for Densely Populated Areas Using LoRaWAN (*Undergraduate Research*)

- Designed and developed a fire detection system prototype using ESP32 and Arduino Uno R3, multi-sensor (DHT22, MQ-2, IR Flame), and LoRaWAN communication.
- Built a *real-time* monitoring dashboard through the ThingsBoard IoT *platform*, achieving 100% *Packet Delivery Ratio* (PDR) in *Line-of-Sight* testing and stable transmission through concrete barriers.
- Integrated automatic Gmail notifications for emergency response, improving early mitigation for densely populated residential areas.
- Created 3D designs using Autodesk AutoCAD and Fusion on the Transmitter side.

Automatic Fire Detection and Extinguishing System on a Robot

- Built an autonomous mobile robot chassis integrating flame detection sensors and a microcontroller (Arduino Uno R3) for navigation and fire-point detection.
- Designed a control system algorithm using the C++ programming language on the Arduino IDE to respond to fire threats in real time and precisely activate the extinguishing actuator mechanism.

Heating Ventilation and Air Conditioning (HVAC)

- Designed the control system architecture and I/O mapping for HVAC system automation using Omron PLC devices.
- Programmed Ladder Diagram control logic using *CX-Programmer software*.
- Designed an interactive and user-friendly HMI (Human Machine Interface) for real-time monitoring and set point adjustment.

CERTIFICATION EXPERIENCE

- **Certificate of Competency: Senior Electrical Equipment Maintenance Technician Level 3 – PT. SENTRA TEKNOLOGI TERAPAN**

Jul 2026

Certified in performing preventive/corrective maintenance on complex electrical systems, troubleshooting, and ensuring compliance with industrial electrical occupational health and safety (K3) standards.

- **ELT Certification (*English Language Test*) - *Language Development IT-PLN***
- **Microsoft Office Specialist Certification - Certiport - A Pearson VUE Business**

Dec 2025

Sep 2023

- **BEES Certification (*Basic English Education Standardization*) - *Language Development IT-PLN***

Feb 2023